MATH 999: Introduction to Widget Theory

Chapter 1 Practice Problems

Exemplar University Fall 2099

Problems

Q1

Compute $\|w\|_W$ for $w = (2, 0, 6, 0, 0, \ldots)$.

Q2

Let $w = (1, 1, 1, \ldots)$. Compute $\|w\|_W$.

Q3

True or false: $\|w\|_W = 0$ implies $w = (0, 0, 0, \ldots)$. Justify.

Q4

Let $w_k^{(n)} = n/(n+k)$. Show $\|w^{(n)}\|_W \leq 1$ for all n and find $\lim_{n\to\infty} \|w^{(n)}\|_W$.

Q5

Prove $\|u + v\|_W \leq \|u\|_W + \|v\|_W$.

Q6

State the Widget Approximation Theorem. What are its hypotheses and conclusions?

Q7

Give an example of a bounded widget sequence that does not converge but has a convergent subsequence.

Q8

Prove the Tail Bound Lemma: if $\|w\|_W \leq M$, then $\sup_{k>K} |w_k| \cdot 2^{-k} \leq M$ for all $K \geq 1$.

Q9

Prove: $\|w^{(n)} - w\|_W \to 0$ implies $w_k^{(n)} \to w_k$ for every k. Is the converse true?

Q10

Prove the Widget Approximation Theorem.

Solutions

Q1

$w = (2, 0, 6, 0, 0, \ldots)$.

$$|w_1| \cdot 2^{-1} = 1,$$
$$|w_3| \cdot 2^{-3} = 6/8 = 3/4.$$

All other terms are 0. So $\|w\|_W = \boxed{1}$.

Q2

$|w_k| \cdot 2^{-k} = 2^{-k}$, maximized at $k = 1$.
$\|w\|_W = 2^{-1} = \boxed{1/2}$.

Q3

True. $\|w\|_W = 0$ means $|w_k| \cdot 2^{-k} \leq 0$ for all k. Since $2^{-k} > 0$, this forces $w_k = 0$ for all k.

Q4

$\frac{n}{n+k} \cdot 2^{-k} \leq 2^{-k} \leq 1/2$ for all $k \geq 1$. So $\|w^{(n)}\|_W \leq 1$.
At $k = 1$: $\frac{n}{n+1} \cdot \frac{1}{2} \to \frac{1}{2}$. For $k \geq 2$: the terms $\to 2^{-k} \leq 1/4 < 1/2$.
$\lim \|w^{(n)}\|_W = \boxed{1/2}$.

Q5

For each k:
$$|u_k + v_k| \cdot 2^{-k} \leq (|u_k| + |v_k|) \cdot 2^{-k} \leq \|u\|_W + \|v\|_W.$$
Taking the sup over k gives $\|u + v\|_W \leq \|u\|_W + \|v\|_W$. $\square$

Q6

Statement: Let $(w^{(n)})$ satisfy $\|w^{(n)}\|_W \leq M$ for all n. Then there exists a subsequence $(w^{(n_j)})$ and $w^* \in W$ with $\|w^{(n_j)} - w^*\|_W \to 0$.
Hypotheses: the sequence is bounded in the widget norm. Conclusion: a norm-convergent subsequence exists.

Q7

$w_k^{(n)} = (-1)^n$ if $k = 1$, and $w_k^{(n)} = 0$ for $k \geq 2$.
Then $\|w^{(n)}\|_W = 1/2$ for all n (bounded), but $w_1^{(n)} = (-1)^n$ does not converge. Taking the subsequence $n = 2j$ gives $w^{(2j)} \to w^*$ with $w_1^* = 1$, $w_k^* = 0$ for $k \geq 2$.

Q8

$\sup_{k>K} |w_k| \cdot 2^{-k}$ is a supremum over a subset of $\{|w_k| \cdot 2^{-k} : k \geq 1\}$, so it is at most the full sup, which is $\|w\|_W \leq M$. $\square$

Q9

($\Rightarrow$) Fix k. Then $|w_k^{(n)} - w_k| \cdot 2^{-k} \leq \|w^{(n)} - w\|_W \to 0$. Since $2^{-k} > 0$: $|w_k^{(n)} - w_k| \leq 2^k \|w^{(n)} - w\|_W \to 0$.
Converse is false. Let $w_k^{(n)} = 2^k$ if $k = n$, and 0 otherwise. Then $w_k^{(n)} \to 0$ for each fixed k. But $\|w^{(n)}\|_W = 2^n \cdot 2^{-n} = 1 \not\to 0$.

Q10

We construct w^* component by component.

Step 1. $(w_1^{(n)})$ is bounded by $2M$. By Bolzano-Weierstrass, extract a subsequence $(w^{(n_j^1)})$ with $w_1^{(n_j^1)} \to w_1^*$.

Step 2. From $(w^{(n_j^1)})$, extract a further subsequence $(w^{(n_j^2)})$ with $w_2^{(n_j^2)} \to w_2^*$.

Step 3. Continue for all k. The diagonal subsequence $w^{(n_j^j)}$ converges in every component.

Step 4. Let $\varepsilon > 0$. Choose K with $M \cdot 2^{-K} < \varepsilon/2$. For j large enough:

$$\|w^{(n_j^j)} - w^*\|_W \leq \max\left(\max_{k \leq K} |w_k^{(n_j^j)} - w_k^*| \cdot 2^{-k},\ M \cdot 2^{-K}\right) < \varepsilon. \quad \square$$